

PharmaOne AI – Workflow & Tech Stack (Updated)

End-to-end workflow, RAG, risk strategy, stack, performance and scope.

1. End-to-End Workflow

Stage	Process
Login	Supabase Auth authenticates Admin or demo user.
Authorization	Load profile + one membership + role; deny inactive/missing.
Input	Upload deviation PDF or text/email.
Extraction	Direct text extraction; OCR only if needed.
Structured AI	Groq/LangGraph extracts validated JSON.
Knowledge	Query embedding retrieves relevant chunks.
Risk	Combine context + deterministic/configured criteria + LLM explanation.
Review	User reviews/edits.
Save	Persist final values + audit event.
Reports	Show saved deviation information.

2. RAG

Embed relatively static reference documents once. Runtime creates a query embedding, retrieves a small top-k set, and passes relevant context to the LLM. RAG supports knowledge retrieval; it does not replace rules or human review.

3. Risk Strategy

Use deterministic checks for straightforward numerical/structural comparisons when approved limits exist. Use the LLM for contextual interpretation and explanation. Demo risk criteria must be labeled configurable/demo criteria, not universal regulatory severity rules. ICH Q9 is methodology guidance, not a universal severity lookup.

4. Tech Stack

Layer	Technology
Frontend	React + Redux
Backend	Python + FastAPI
AI	LangGraph + Groq
Database	Supabase PostgreSQL
Vector search	PostgreSQL/pgvector or compatible vector storage
Auth	Supabase Auth
Storage	Supabase Storage
Migrations	Alembic
Local	Docker Compose
Deployment	Vercel + Render + Supabase

5. Performance

Skip OCR for text PDFs. Precompute reference embeddings. Cache extraction/OCR where safe. Keep retrieval top-k small. Use structured LLM output. Parallelize independent work where safe.

Track p50/p95/p99, error rate and stage timings. Suggested engineering targets: non-AI API p95 <500 ms; normal AI workflow p95 roughly <10–15 seconds depending on document size and external service latency.

6. Failure Handling

Invalid PDF → friendly error. OCR/LLM failure → controlled retry/error. Invalid JSON → validate/repair or reject.

Database failure → do not report success.

7. Scope

Do not build public registration, company switching, multiple-company users, enterprise SSO, LDAP, SCIM, billing, complex RBAC or advanced organization management in this version.